OVP Guide to Using Processor Models

Model specific information for ARM_Cortex-M33F

Imperas Software Limited
Imperas Buildings, North Weston
Thame, Oxfordshire, OX9 2HA, U.K.
docs@imperas.com

Author	Imperas Software Limited
Version	20250716.0
Filename	OVP_Model_Specific_Information_armm_Cortex-M33F.pdf
Created	16 July 2025
Status	OVP Standard Release

Model Release Status

This model is released as part of OVP releases and is included in OVPworld packages. Please visit OVPworld.org.

Contents

Chapter 1

Overview

This document provides the details of an OVP Fast Processor Model variant.

OVP Fast Processor Models are written in C and provide a C API for use in C based platforms. The models also provide a native interface for use in SystemC TLM2 platforms.

The models are written using the OVP VMI API that provides a Virtual Machine Interface that defines the behavior of the processor. The VMI API makes a clear line between model and simulator allowing very good optimization and world class high speed performance. Most models are provided as a binary shared object and also as source. This allows the download and use of the model binary or the use of the source to explore and modify the model.

The models are run through an extensive QA and regression testing process and most model families are validated using technology provided by the processor IP owners. There is a companion document (OVP Guide to Using Processor Models) which explains the general concepts of OVP Fast Processor Models and their use. It is downloadable from the OVPworld website documentation pages.

1.1 Description

ARMM Processor Model

1.2 Licensing

Usage of binary model under license governing simulator usage.

Note that for models of ARM CPUs the license includes the following terms:

Licensee is granted a non-exclusive, worldwide, non-transferable, revocable licence to:

If no source is being provided to the Licensee: use and copy only (no modifications rights are granted) the model for the sole purpose of designing, developing, analyzing, debugging, testing, verifying, validating and optimizing software which: (a) (i) is for ARM based systems; and (ii) does not incorporate the ARM Models or any part thereof; and (b) such ARM Models may not be used to emulate an ARM based system to run application software in a production or live environment.

If source code is being provided to the Licensee: use, copy and modify the model for the sole purpose of designing, developing, analyzing, debugging, testing, verifying, validating and optimizing software which: (a) (i) is for ARM based systems; and (ii) does not incorporate the ARM Models or any part thereof; and (b) such ARM Models may not be used to emulate an ARM based system to run application software in a production or live environment.

In the case of any Licensee who is either or both an academic or educational institution the purposes shall be limited to internal use.

Except to the extent that such activity is permitted by applicable law, Licensee shall not reverse engineer, decompile, or disassemble this model. If this model was provided to Licensee in Europe, Licensee shall not reverse engineer, decompile or disassemble the Model for the purposes of error correction.

The License agreement does not entitle Licensee to manufacture in silicon any product based on this model.

The License agreement does not entitle Licensee to use this model for evaluating the validity of any ARM patent.

The License agreement does not entitle Licensee to use the model to emulate an ARM based system to run application software in a production or live environment.

Source of model available under separate Imperas Software License Agreement.

1.3 Limitations

Performance Monitors are not implemented.

Debug Extension and related blocks are not implemented.

1.4 Verification

Models have been extensively tested by Imperas. ARM Cortex-M models have been successfully used by customers to simulate the Micrium uC/OS-II kernel and FreeRTOS.

1.5 Features

The model is configured with 16 interrupts and 3 priority bits (use override_numInterrupts and override_priorityBits parameters to change these).

Thumb-2 instructions are supported.

MPU is present. Use parameter override_MPU_TYPE to disable it or change the number of MPU regions if required.

SysTick timer is present. Use parameter SysTickPresent to disable it if required.

FPU extension is present. Use parameter override_MVFR0 to disable it if required.

DSP extension is present. Use parameter override_InstructionAttributes3 to disable it if required.

Bit-band region is not present. Use parameter BitBandPresent to enable it if required.

SAU is present. Use parameter override_SAU_TYPE to disable it or change the number of SAU regions if required.

SIE-200 IDAU not is present. Use parameter SIE200IDAUPresent to enable it if required.

A 20-bit PPB bus port can be connected. If it is, then any loads or stores to address range 0xE0040000:0xE00FFFFF will be visible on this bus at address range 0x00040000:0x000FFFFF.

1.6 Unpredictable Behavior

Many instruction behaviors are described in the ARM ARM as CONSTRAINED UNPREDICTABLE. This section describes how such situations are handled by this model.

1.6.1 Equal Target Registers

Some instructions allow the specification of two target registers (for example, double-width SMULL, or some VMOV variants), and such instructions are CONSTRAINED UNPREDICTABLE if the same target register is specified in both positions. In this model, such instructions are treated as UNDEFINED.

1.6.2 Floating Point Load/Store Multiple Lists

Instructions that load or store a list of floating point registers (e.g. VSTM, VLDM, VPUSH, VPOP) are CONSTRAINED UNPREDICTABLE if either the uppermost register in the specified range is greater than 32 or (for 64-bit registers) if more than 16 registers are specified. In this model, such instructions are treated as UNDEFINED.

1.6.3 If-Then (IT) Block Constraints

Where the behavior of an instruction in an if-then (IT) block is described as CONSTRAINED UNPREDICTABLE, this model treats that instruction as UNDEFINED.

1.6.4 Use of R13

Use of R13 is described as CONSTRAINED UNPREDICTABLE in many circumstances. This model allows R13 to be used like any other GPR.

1.6.5 Use of R15

Use of R15 is described as CONSTRAINED UNPREDICTABLE in many circumstances. This model allows such use to be configured using the parameter "unpredictableR15" as follows:

Value "undefined": any reference to R15 in such a situation is treated as UNDEFINED;

Value "nop": any reference to R15 in such a situation causes the instruction to be treated as a NOP;

Value "raz_wi": any reference to R15 in such a situation causes the instruction to be treated as a RAZ/WI (that is, R15 is read as zero and write-ignored);

Value "execute": any reference to R15 in such a situation is executed using the current value of R15 on read, and writes to R15 are allowed.

Value "assert": any reference to R15 in such a situation causes the simulation to halt with an assertion message (allowing any such unpredictable uses to be easily identified).

In this variant, the default value of "unpredictableR15" is "execute".

1.7 Integration Support

This model implements a number of non-architectural pseudo-registers and other features to facilitate integration.

1.7.1 Memory Transaction Query

Two registers are intended for use within memory callback functions to provide additional information about the current memory access. Register executionPri indicates the current processor execution priority. Register atomicType indicates whether the current instruction has any special atomic constraints (0 indicates no constraint, 1 indicates atomic, 2 indicates exclusive access).

1.7.2 Halt Reason Introspection

An artifact register HaltReason can be read to determine the reason or reasons that a processor is halted. This register is a bitfield, with the following encoding: bit 0 indicates the processor has executed a wait-for-event (WFE) instruction; bit 1 indicates the processor has executed a wait-for-interrupt (WFI) instruction; bit 2 indicates the processor has entered lockup state; and bit 3 indicates the processor is held in reset.

1.7.3 Automatic WFI Wake Up

A simulation environment can cause a core to wake up from the WFI state by writing to the artifact register WakeWFINow.

Chapter 2

Configuration

2.1 Location

This model's VLNV is arm.ovpworld.org/processor/armm/1.0.
The model source is usually at:
$IMPERAS_HOME/ImperasLib/source/arm.ovpworld.org/processor/armm/1.0
The model binary is usually at:
$IMPERAS_HOME/lib/$IMPERAS_ARCH/ImperasLib/arm.ovpworld.org/processor/armm/1.0

2.2 GDB Path

The default GDB for this model is: $IMPERAS_HOME/lib/$IMPERAS_ARCH/gdb/arm-none-eabi-gdb.

2.3 Semi-Host Library

The default semi-host library file is arm.ovpworld.org/semihosting/armNewlib/1.0

2.4 Processor Endian-ness

This is a LITTLE endian model.

2.5 QuantumLeap Support

This processor is qualified to run in a QuantumLeap enabled simulator.

2.6 Processor ELF code

The ELF code supported by this model is: 0x28.

Chapter 3

All Variants in this model

This model has these variants

Variant	Description
ARMv6-M	
ARMv7-M	
Cortex-M0	
Cortex-M0plus	
Cortex-M1	
Cortex-M3	
Cortex-M4	
Cortex-M4F	
Cortex-M7	
Cortex-M7F	
Cortex-M23	
Cortex-M33	
Cortex-M33F	(described in this document)
Cortex-M55	
Cortex-M55F	

Table 3.1: All Variants in this model

Chapter 4

Bus Master Ports

This model has these bus master ports.

Name	min	max	Connect?	Description
INSTRUCTION	32	33	mandatory	
DATA	32	33	optional	
PPB	20	20	optional	PPB interface

Table 4.1: Bus Master Ports

Chapter 5

Bus Slave Ports

This model has no bus slave ports.

Chapter 6

Net Ports

This model has these net ports.

Name	Type	Connect?	Description
sysResetReq	output	optional	Indicates that a reset is required
intISS	output	optional	Indicates that an interrupt service has started
eventOut	output	optional	EVENTO output signal (SEV)
lockup	output	optional	Indicates fatal lockup state entered
haltReason	output	optional	Indicates why core is halted
currPri	output	optional	Current execution priority
currNS	output	optional	Current security state
intnum	output	optional	Exception number of current execution context
int	input	optional	Raise or lower interrupt by index
reset	input	optional	RESET (active high)
cpuWait	input	optional	CPUWAIT (active high at reset only)
nmi	input	optional	NMI (posedge-triggered)
eventIn	input	optional	EVENTI input (posedge-triggered)
initnsvtor	input	optional	Non-secure VTOR reset configuration
initsvtor	input	optional	Secure VTOR reset configuration
fpxxc	output	optional	Cumulative floating point exception flags
LOCKSAU	input	optional	LOCKSAU system register lock
int0	input	optional	Scalar interrupt input 0
int1	input	optional	Scalar interrupt input 1
int2	input	optional	Scalar interrupt input 2
int3	input	optional	Scalar interrupt input 3
int4	input	optional	Scalar interrupt input 4
int5	input	optional	Scalar interrupt input 5
int6	input	optional	Scalar interrupt input 6
int7	input	optional	Scalar interrupt input 7
int8	input	optional	Scalar interrupt input 8
int9	input	optional	Scalar interrupt input 9
int10	input	optional	Scalar interrupt input 10

int11	input	optional	Scalar interrupt input 11
int12	input	optional	Scalar interrupt input 12
int13	input	optional	Scalar interrupt input 13
int14	input	optional	Scalar interrupt input 14
int15	input	optional	Scalar interrupt input 15

Table 6.1: Net Ports

Chapter 7

FIFO Ports

This model has no FIFO ports.

Chapter 8

Formal Parameters

Name	Type	Description
verbose	Boolean	Specify verbosity of output
showHiddenRegs	Boolean	Show hidden registers during register tracing
UAL	Boolean	Disassemble using UAL syntax
enableVFPAtReset	Boolean	Enable vector floating point (SIMD and VFP) instructions at reset. (Enables cp10/11 in CPACR and sets FPEXC.EN)
compatibility	Enumeration	Specify compatibility mode
	ISA	
	gdb	
	nopBKPT	
unpredictableR15	Enumeration	Specify behavior for UNPREDICTABLE uses of R15
	undefined	
	nop	
	raz_wi	
	execute	
	assert	
override_debugMask	Uns32	Specifies debug mask, enabling debug output for model components
instructionEndian	Endian	The architecture specifies that instruction fetch is always little endian; this attribute allows the defined instruction endianness to be overridden if required
resetAtTime0	Boolean	Reset the model at time=0 (default=1)
SysTickPresent	Uns32	Specify number of SysTick timers present
BitBandPresent	Boolean	Specify presence of bit-band region
SIE200IDAUPresent	Boolean	Specify presence of SIE-200 Implementation-Defined Attribution Unit
override_PFR0	Uns32	Override ID_PFR0 register
override_PFR1	Uns32	Override ID_PFR1 register
override_DFR0	Uns32	Override ID_DFR0 register
override_AFR0	Uns32	Override ID_AFR0 register
override_MMFR0	Uns32	Override ID_MMFR0 register
override_MMFR1	Uns32	Override ID_MMFR1 register
override_MMFR2	Uns32	Override ID_MMFR2 register
override_MMFR3	Uns32	Override ID_MMFR3 register
override_ISAR0	Uns32	Override ID_ISAR0 register
override_ISAR1	Uns32	Override ID_ISAR1 register
override_ISAR2	Uns32	Override ID_ISAR2 register
override_ISAR3	Uns32	Override ID_ISAR3 register
override_ISAR4	Uns32	Override ID_ISAR4 register
override_ISAR5	Uns32	Override ID_ISAR5 register

override_MVFR0	Uns32	Override ID_MVFR0 register
override_MVFR1	Uns32	Override ID_MVFR1 register
override_MVFR2	Uns32	Override ID_MVFR2 register
override_CPUID	Uns32	Override system CPUID register
override_MPU_TYPE	Uns32	Override system MPU_TYPE register
override_SAU_TYPE	Uns32	Override system SAU_TYPE register
override_VTOR	Uns32	Override VTOR register reset value
override_CCSIDR_1I	Uns32	Override CCSIDR (level 1 instruction)
override_CCSIDR_1D	Uns32	Override CCSIDR (level 1 data)
override_CCSIDR_2I	Uns32	Override CCSIDR (level 2 instruction)
override_CCSIDR_2D	Uns32	Override CCSIDR (level 2 data)
override_CCSIDR_3I	Uns32	Override CCSIDR (level 3 instruction)
override_CCSIDR_3D	Uns32	Override CCSIDR (level 3 data)
override_CCSIDR_4I	Uns32	Override CCSIDR (level 4 instruction)
override_CCSIDR_4D	Uns32	Override CCSIDR (level 4 data)
override_CCSIDR_5I	Uns32	Override CCSIDR (level 5 instruction)
override_CCSIDR_5D	Uns32	Override CCSIDR (level 5 data)
override_CCSIDR_6I	Uns32	Override CCSIDR (level 6 instruction)
override_CCSIDR_6D	Uns32	Override CCSIDR (level 6 data)
override_CCSIDR_7I	Uns32	Override CCSIDR (level 7 instruction)
override_CCSIDR_7D	Uns32	Override CCSIDR (level 7 data)
override_deviceStrongAligned	Boolean	Force accesses to Device and Strongly Ordered regions to be aligned
override_STRoffsetPC12	Uns32	Specifies that STR/STR of PC should do so with 12:byte offset from the current instruction (if 1), otherwise an 8:byte offset is used
override_ERG	Uns32	Specifies exclusive reservation granule
override_priorityBits	Uns32	Specifies number of priority bits in BASEPRI etc (1-8, default is 3)
override_numInterrupts	Uns32	Specifies number of external interrupt lines
override_InstructionAttributes0	Uns32	Override ID_ISAR0 register (deprecated, use override_ISAR0)
override_InstructionAttributes1	Uns32	Override ID_ISAR1 register (deprecated, use override_ISAR1)
override_InstructionAttributes2	Uns32	Override ID_ISAR2 register (deprecated, use override_ISAR2)
override_InstructionAttributes3	Uns32	Override ID_ISAR3 register (deprecated, use override_ISAR3)
override_InstructionAttributes4	Uns32	Override ID_ISAR4 register (deprecated, use override_ISAR4)
override_InstructionAttributes5	Uns32	Override ID_ISAR5 register (deprecated, use override_ISAR5)

Table 8.1: Parameters

8.1 Parameter values and limits

These are the formal parameter limits and actual parameter values

Name	Min	Max	Default	Actual
(Others)				
variant				Cortex-M33F
verbose			t	t
showHiddenRegs			f	f
UAL			t	t
enableVFPAtReset			f	f
compatibility			ISA	ISA
unpredictableR15			execute	execute

override_debugMask	0	0xffffffff	0	0
endian			none	none
instructionEndian			none	none
resetAtTime0			t	t
SysTickPresent	1	2	1	1
BitBandPresent			f	f
SIE200IDAUPresent			f	f
override_PFR0	0	0xffffffff	48	48
override_PFR1	0	0xffffffff	0x210	0x210
override_DFR0	0	0xffffffff	0	0
override_AFR0	0	0xffffffff	0	0
override_MMFR0	0	0xffffffff	0x101f40	0x101f40
override_MMFR1	0	0xffffffff	0	0
override_MMFR2	0	0xffffffff	0x1000000	0x1000000
override_MMFR3	0	0xffffffff	0	0
override_ISAR0	0	0xffffffff	0x1101110	0x1101110
override_ISAR1	0	0xffffffff	0x2212000	0x2212000
override_ISAR2	0	0xffffffff	0x20232232	0x20232232
override_ISAR3	0	0xffffffff	0x1111131	0x1111131
override_ISAR4	0	0xffffffff	0x1310132	0x1310132
override_ISAR5	0	0xffffffff	0	0
override_MVFR0	0	0xffffffff	0x10110221	0x10110221
override_MVFR1	0	0xffffffff	0x12000011	0x12000011
override_MVFR2	0	0xffffffff	64	64
override_CPUID	0	0xffffffff	0x410fd214	0x410fd214
override_MPU_TYPE	0	0xffffffff	0x800	0x800
override_SAU_TYPE	0	0xffffffff	4	4
override_VTOR	0	0xffffffff	0	0
override_CCSIDR_1I	0	0xffffffff	0xf03fe019	0xf03fe019
override_CCSIDR_1D	0	0xffffffff	0xf03fe019	0xf03fe019
override_CCSIDR_2I	0	0xffffffff	0	0
override_CCSIDR_2D	0	0xffffffff	0	0
override_CCSIDR_3I	0	0xffffffff	0	0
override_CCSIDR_3D	0	0xffffffff	0	0
override_CCSIDR_4I	0	0xffffffff	0	0
override_CCSIDR_4D	0	0xffffffff	0	0
override_CCSIDR_5I	0	0xffffffff	0	0
override_CCSIDR_5D	0	0xffffffff	0	0
override_CCSIDR_6I	0	0xffffffff	0	0
override_CCSIDR_6D	0	0xffffffff	0	0
override_CCSIDR_7I	0	0xffffffff	0	0
override_CCSIDR_7D	0	0xffffffff	0	0
override_deviceStrongAligned			t	t
override_STRoffsetPC12	0	1	0	0
override_ERG	0	0x400	0	0

override_priorityBits	1	8	3	3
override_numInterrupts	0	0x1f0	16	16
override_InstructionAttributes0	0	0xffffffff	0x1101110	0x1101110
override_InstructionAttributes1	0	0xffffffff	0x2212000	0x2212000
override_InstructionAttributes2	0	0xffffffff	0x20232232	0x20232232
override_InstructionAttributes3	0	0xffffffff	0x1111131	0x1111131
override_InstructionAttributes4	0	0xffffffff	0x1310132	0x1310132
override_InstructionAttributes5	0	0xffffffff	0	0

Table 8.2: Parameter values and limits

Chapter 9

Execution Modes

Mode	Code
Thread_NS	0
Handler_NS	1
Thread_S	2
Handler_S	3

Table 9.1: Modes implemented in this processor

Chapter 10

Exceptions

Exception	Code
None	0
Reset	1
NMI	2
HardFault	3
MemManage	4
BusFault	5
UsageFault	6
SecureFault	7
SVCall	11
DebugMonitor	12
PendSV	14
SysTick	15
ExternalInt000	16
ExternalInt001	17
ExternalInt002	18
ExternalInt003	19
ExternalInt004	20
ExternalInt005	21
ExternalInt006	22
ExternalInt007	23
ExternalInt008	24
ExternalInt009	25
ExternalInt00a	26
ExternalInt00b	27
ExternalInt00c	28
ExternalInt00d	29
ExternalInt00e	30
ExternalInt00f	31

Table 10.1: Exceptions implemented by this processor

Chapter 11

Hierarchy of the model

A CPU core may be configured to instance many processors of a Symmetrical Multi Processor (SMP). A CPU core may also have sub elements within a processor, for example hardware threading blocks.
OVP processor models can be written to include SMP blocks and to have many levels of hierarchy. Some OVP CPU models may have a fixed hierarchy, and some may be configured by settings in a configuration register. Please see the register definitions of this model.
This model documentation shows the settings and hierarchy of the default settings for this model variant.

11.1 Level 1

This level in the model hierarchy has 3 commands.
This level in the model hierarchy has 7 register groups:

Group name	Registers
Core	16
Control	26
System	90
System_secure	36
System_non_secure	36
VFP	17
Integration_support	5

Table 11.1: Register groups

This level in the model hierarchy has no children.

Chapter 12

Model Commands

A Processor model can implement one or more **Model Commands** available to be invoked from the simulator command line, from the OP API or from the Imperas Multiprocessor Debugger.

12.1 Level 1

12.1.1 debugflags

show or modify the processor debug flags

Argument	Type	Description
-get	Boolean	print current processor flags value
-mask	Boolean	print valid debug flag bits
-set	Int32	new processor flags (only flags 0x0000008c can be modified)

Table 12.1: debugflags command arguments

12.1.2 isync

specify instruction address range for synchronous execution

Argument	Type	Description
-addresshi	Uns64	end address of synchronous execution range
-addresslo	Uns64	start address of synchronous execution range

Table 12.2: isync command arguments

12.1.3 itrace

enable or disable instruction tracing

Argument	Type	Description
-access	String	show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)

-after	Uns64	apply after this many instructions
-enable	Boolean	enable instruction tracing
-full	Boolean	turn on all trace features
-instructioncount	Boolean	include the instruction number in each trace
-memory	String	(Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-mode	Boolean	show processor mode changes
-off	Boolean	disable instruction tracing
-on	Boolean	enable instruction tracing
-processorname	Boolean	Include processor name in all trace lines
-registerchange	Boolean	show registers changed by this instruction
-registers	Boolean	show registers after each trace

Table 12.3: itrace command arguments

Chapter 13

Registers

13.1 Level 1

13.1.1 Core

Registers at level:1, group:Core

Name	Bits	Initial-Hex	RW	Description
r0	32	0	rw	
r1	32	0	rw	
r2	32	0	rw	
r3	32	0	rw	
r4	32	0	rw	
r5	32	0	rw	
r6	32	0	rw	
r7	32	0	rw	
r8	32	0	rw	
r9	32	0	rw	
r10	32	0	rw	
r11	32	0	rw	frame pointer
r12	32	0	rw	
sp	32	0	rw	stack pointer
lr	32	0	rw	
pc	32	0	rw	program counter

Table 13.1: Registers at level 1, group:Core

13.1.2 Control

Registers at level:1, group:Control

Name	Bits	Initial-Hex	RW	Description
fps	32	0	rw	archaic FPSCR view (for gdb)
cpsr	32	0	rw	xPSR register. Includes APSR, IPSR and EPSR
control	32	0	rw	
primask	32	0	rw	
faultmask	32	0	rw	
basepri	32	0	rw	
control_S	32	0	rw	
primask_S	32	0	rw	
faultmask_S	32	0	rw	

basepri_S	32	0	rw	
control_NS	32	0	rw	
primask_NS	32	0	rw	
faultmask_NS	32	0	rw	
basepri_NS	32	0	rw	
sp_process	32	0	rw	stack pointer
sp_process_S	32	0	rw	stack pointer
sp_process_NS	32	0	rw	stack pointer
msplim	32	0	rw	
msplim_S	32	0	rw	
msplim_NS	32	0	rw	
psplim	32	0	rw	
psplim_S	32	0	rw	
psplim_NS	32	0	rw	
sp_main	32	0	rw	stack pointer
sp_main_S	32	0	rw	stack pointer
sp_main_NS	32	0	rw	stack pointer

Table 13.2: Registers at level 1, group:Control

13.1.3 System

Registers at level:1, group:System

Name	Bits	Initial-Hex	RW	Description
ICTR	32	0	rw	0xe000e004: Interrupt Controller Type
ACTLR	32	0	rw	0xe000e008: Auxiliary Control
CPPWR	32	0	rw	0xe000e00c: Coprocessor Power Control
SYST_CSR	32	4	rw	0xe000e010: SysTick Control and Status
SYST_RVR	32	0	rw	0xe000e014: SysTick Reload Value
SYST_CVR	32	0	rw	0xe000e018: SysTick Current Value
SYST_CALIB	32	0	rw	0xe000e01c: SysTick Calibration Value
NVIC_ISER0	32	0	rw	0xe000e100: Interrupt Set Enable 0
NVIC_ICER0	32	0	rw	0xe000e180: Interrupt Clear Enable 0
NVIC_ISPR0	32	0	rw	0xe000e200: Interrupt Set Pending 0
NVIC_ICPR0	32	0	rw	0xe000e280: Interrupt Clear Pending 0
NVIC_IABR0	32	0	r-	0xe000e300: Interrupt Active Bit 0
NVIC_ITNS0	32	0	rw	0xe000e380: Interrupt Target Non-Secure 0
NVIC_IPR0	32	0	rw	0xe000e400: Interrupt Priority 0
NVIC_IPR1	32	0	rw	0xe000e404: Interrupt Priority 1
NVIC_IPR2	32	0	rw	0xe000e408: Interrupt Priority 2
NVIC_IPR3	32	0	rw	0xe000e40c: Interrupt Priority 3
CPUID	32	410fd214	r-	0xe000ed00: CPUID Base
ICSR	32	1000	rw	0xe000ed04: Interrupt Control and State
VTOR	32	0	rw	0xe000ed08: Vector Table Offset
AIRCR	32	fa050000	rw	0xe000ed0c: Application Interrupt and Reset Control
SCR	32	0	rw	0xe000ed10: System Control
CCR	32	201	rw	0xe000ed14: Configuration and Control
SHPR1	32	0	rw	0xe000ed18: System Handler Priority 1
SHPR2	32	0	rw	0xe000ed1c: System Handler Priority 2
SHPR3	32	0	rw	0xe000ed20: System Handler Priority 3
SHCSR	32	0	rw	0xe000ed24: System Handler Control and State
CFSR	32	0	rw	0xe000ed28: Configurable Fault Status
HFSR	32	0	rw	0xe000ed2c: HardFault Status
DFSR	32	0	rw	0xe000ed30: Debug Fault Status Register

MMAR	32	0	rw	0xe000ed34: MemManage Fault Address
BFAR	32	0	rw	0xe000ed38: BusFault Address
AFSR	32	0	rw	0xe000ed3c: Auxiliary Fault Status
ID_PFR0	32	30	rw	0xe000ed40: Processor Feature 0
ID_PFR1	32	210	rw	0xe000ed44: Processor Feature 1
ID_DFR0	32	0	rw	0xe000ed48: Debug Feature 0
ID_AFR0	32	0	rw	0xe000ed4c: Auxiliary Feature 0
ID_MMFR0	32	101f40	rw	0xe000ed50: Memory Model Feature 0
ID_MMFR1	32	0	rw	0xe000ed54: Memory Model Feature 1
ID_MMFR2	32	1000000	rw	0xe000ed58: Memory Model Feature 2
ID_MMFR3	32	0	rw	0xe000ed5c: Memory Model Feature 3
ID_ISAR0	32	1101110	rw	0xe000ed60: Instruction Set Attributes 0
ID_ISAR1	32	2212000	rw	0xe000ed64: Instruction Set Attributes 1
ID_ISAR2	32	20232232	rw	0xe000ed68: Instruction Set Attributes 2
ID_ISAR3	32	1111131	rw	0xe000ed6c: Instruction Set Attributes 3
ID_ISAR4	32	1310132	rw	0xe000ed70: Instruction Set Attributes 4
ID_ISAR5	32	0	rw	0xe000ed74: Instruction Set Attributes 5
CLIDR	32	9000003	r-	0xe000ed78: Cache Level ID
CTR	32	8003c003	r-	0xe000ed7c: Cache Type
CCSIDR	32	f03fe019	r-	0xe000ed80: Cache Size ID
CSSELR	32	0	rw	0xe000ed84: Cache Size Selection
CPACR	32	0	rw	0xe000ed88: Coprocessor Access Control
NSACR	32	0	rw	0xe000ed8c: Non-Secure Access Control Register
MPU_TYPE	32	800	rw	0xe000ed90: MPU Type
MPU_CONTROL	32	0	rw	0xe000ed94: MPU Control
MPU_RNR	32	0	rw	0xe000ed98: MPU Region Number
MPU_RBAR	32	0	rw	0xe000ed9c: MPU Region Base Address
MPU_RLAR	32	0	rw	0xe000eda0: MPU Region Limit Address Register
MPU_RBAR_A1	32	0	rw	0xe000eda4: MPU Region Base Address Alias 1
MPU_RLAR_A1	32	0	rw	0xe000eda8: MPU Region Limit Address Register Alias 1
MPU_RBAR_A2	32	0	rw	0xe000edac: MPU Region Base Address Alias 2
MPU_RLAR_A2	32	0	rw	0xe000edb0: MPU Region Limit Address Register Alias 2
MPU_RBAR_A3	32	0	rw	0xe000edb4: MPU Region Base Address Alias 3
MPU_RLAR_A3	32	0	rw	0xe000edb8: MPU Region Limit Address Register Alias 3
MPU_MAIR0	32	0	rw	0xe000edc0: MPU Memory Attribute Indirection Register 0
MPU_MAIR1	32	0	rw	0xe000edc4: MPU Memory Attribute Indirection Register 1
SAU_CTRL	32	0	rw	0xe000edd0: SAU Control Register
SAU_TYPE	32	4	rw	0xe000edd4: SAU Type Register
SAU_RNR	32	0	rw	0xe000edd8: SAU Region Number Register
SAU_RBAR	32	0	rw	0xe000eddc: SAU Region Base Address Register
SAU_RLAR	32	0	rw	0xe000ede0: SAU Region Limit Address Register
SFSR	32	0	rw	0xe000ede4: Secure Fault Status Register
SFAR	32	0	rw	0xe000ede8: Secure Fault Address Register
DEMCR	32	100000	rw	0xe000edfc: Debug Exception and Monitor Control
STIR	32	-	-w	0xe000ef00: Software Triggered Interrupt
FPCCR	32	c0000004	rw	0xe000ef34: Floating Point Context Control
FPCAR	32	0	rw	0xe000ef38: Floating Point Context Address
FPDSCR	32	0	rw	0xe000ef3c: Floating Point Default Status Control
MVFR0	32	10110221	r-	0xe000ef40: Media and VFP Feature 0
MVFR1	32	12000011	r-	0xe000ef44: Media and VFP Feature 1
MVFR2	32	40	r-	0xe000ef48: Media and VFP Feature 2
ICIALLU	32	-	-w	0xe000ef50: Instruction Cache Invalidate All to PoU
ICIMVAU	32	-	-w	0xe000ef58: Instruction Cache Invalidate by Address to PoU
DCIMVAC	32	-	-w	0xe000ef5c: Data Cache Invalidate by Address to PoC
DCISW	32	-	-w	0xe000ef60: Data Cache Invalidate by Set/Way
DCCMVAU	32	-	-w	0xe000ef64: Data Cache Invalidate by Address to PoU

DCCMVAC	32	-	-w	0xe000ef68: Data Cache Clean by Address to PoC
DCCSW	32	-	-w	0xe000ef6c: Data Cache Clean by Set/Way
DCCIMVAC	32	-	-w	0xe000ef70: Data Cache Clean and Invalidate by Address to PoC
DCCISW	32	-	-w	0xe000ef74: Data Cache Clean and Invalidate by Set/Way

Table 13.3: Registers at level 1, group:System

13.1.4 System_secure

Registers at level:1, group:System_secure

Name	Bits	Initial-Hex	RW	Description
ACTLR_S	32	0	rw	0xe000e008: Auxiliary Control
SYST_CSR_S	32	4	rw	0xe000e010: SysTick Control and Status
SYST_RVR_S	32	0	rw	0xe000e014: SysTick Reload Value
SYST_CVR_S	32	0	rw	0xe000e018: SysTick Current Value
SYST_CALIB_S	32	0	rw	0xe000e01c: SysTick Calibration Value
ICSR_S	32	1000	rw	0xe000ed04: Interrupt Control and State
VTOR_S	32	0	rw	0xe000ed08: Vector Table Offset
AIRCR_S	32	fa050000	rw	0xe000ed0c: Application Interrupt and Reset Control
SCR_S	32	0	rw	0xe000ed10: System Control
CCR_S	32	201	rw	0xe000ed14: Configuration and Control
SHPR1_S	32	0	rw	0xe000ed18: System Handler Priority 1
SHPR2_S	32	0	rw	0xe000ed1c: System Handler Priority 2
SHPR3_S	32	0	rw	0xe000ed20: System Handler Priority 3
SHCSR_S	32	0	rw	0xe000ed24: System Handler Control and State
CFSR_S	32	0	rw	0xe000ed28: Configurable Fault Status
MMAR_S	32	0	rw	0xe000ed34: MemManage Fault Address
CTR_S	32	8003c003	r-	0xe000ed7c: Cache Type
CCSIDR_S	32	f03fe019	r-	0xe000ed80: Cache Size ID
CSSELR_S	32	0	rw	0xe000ed84: Cache Size Selection
CPACR_S	32	0	rw	0xe000ed88: Coprocessor Access Control
MPU_TYPE_S	32	800	rw	0xe000ed90: MPU Type
MPU_CONTROL_S	32	0	rw	0xe000ed94: MPU Control
MPU_RNR_S	32	0	rw	0xe000ed98: MPU Region Number
MPU_RBAR_S	32	0	rw	0xe000ed9c: MPU Region Base Address
MPU_RLAR_S	32	0	rw	0xe000eda0: MPU Region Limit Address Register
MPU_RBAR_A1_S	32	0	rw	0xe000eda4: MPU Region Base Address Alias 1
MPU_RLAR_A1_S	32	0	rw	0xe000eda8: MPU Region Limit Address Register Alias 1
MPU_RBAR_A2_S	32	0	rw	0xe000edac: MPU Region Base Address Alias 2
MPU_RLAR_A2_S	32	0	rw	0xe000edb0: MPU Region Limit Address Register Alias 2
MPU_RBAR_A3_S	32	0	rw	0xe000edb4: MPU Region Base Address Alias 3
MPU_RLAR_A3_S	32	0	rw	0xe000edb8: MPU Region Limit Address Register Alias 3
MPU_MAIR0_S	32	0	rw	0xe000edc0: MPU Memory Attribute Indirection Register 0
MPU_MAIR1_S	32	0	rw	0xe000edc4: MPU Memory Attribute Indirection Register 1
FPCCR_S	32	c0000004	rw	0xe000ef34: Floating Point Context Control
FPCAR_S	32	0	rw	0xe000ef38: Floating Point Context Address
FPDSCR_S	32	0	rw	0xe000ef3c: Floating Point Default Status Control

Table 13.4: Registers at level 1, group:System_secure

13.1.5 System_non_secure

Registers at level:1, group:System_non_secure

Name	Bits	Initial-Hex	RW	Description
ACTLR_NS	32	0	rw	0xe000e008: Auxiliary Control
SYST_CSR_NS	32	4	rw	0xe000e010: SysTick Control and Status
SYST_RVR_NS	32	0	rw	0xe000e014: SysTick Reload Value
SYST_CVR_NS	32	0	rw	0xe000e018: SysTick Current Value
SYST_CALIB_NS	32	0	rw	0xe000e01c: SysTick Calibration Value
ICSR_NS	32	1000	rw	0xe000ed04: Interrupt Control and State
VTOR_NS	32	0	rw	0xe000ed08: Vector Table Offset
AIRCR_NS	32	fa050000	rw	0xe000ed0c: Application Interrupt and Reset Control
SCR_NS	32	0	rw	0xe000ed10: System Control
CCR_NS	32	201	rw	0xe000ed14: Configuration and Control
SHPR1_NS	32	0	rw	0xe000ed18: System Handler Priority 1
SHPR2_NS	32	0	rw	0xe000ed1c: System Handler Priority 2
SHPR3_NS	32	0	rw	0xe000ed20: System Handler Priority 3
SHCSR_NS	32	0	rw	0xe000ed24: System Handler Control and State
CFSR_NS	32	0	rw	0xe000ed28: Configurable Fault Status
MMAR_NS	32	0	rw	0xe000ed34: MemManage Fault Address
CTR_NS	32	8003c003	r-	0xe000ed7c: Cache Type
CCSIDR_NS	32	f03fe019	r-	0xe000ed80: Cache Size ID
CSSELR_NS	32	0	rw	0xe000ed84: Cache Size Selection
CPACR_NS	32	0	rw	0xe000ed88: Coprocessor Access Control
MPU_TYPE_NS	32	800	rw	0xe000ed90: MPU Type
MPU_CONTROL_NS	32	0	rw	0xe000ed94: MPU Control
MPU_RNR_NS	32	0	rw	0xe000ed98: MPU Region Number
MPU_RBAR_NS	32	0	rw	0xe000ed9c: MPU Region Base Address
MPU_RLAR_NS	32	0	rw	0xe000eda0: MPU Region Limit Address Register
MPU_RBAR_A1_NS	32	0	rw	0xe000eda4: MPU Region Base Address Alias 1
MPU_RLAR_A1_NS	32	0	rw	0xe000eda8: MPU Region Limit Address Register Alias 1
MPU_RBAR_A2_NS	32	0	rw	0xe000edac: MPU Region Base Address Alias 2
MPU_RLAR_A2_NS	32	0	rw	0xe000edb0: MPU Region Limit Address Register Alias 2
MPU_RBAR_A3_NS	32	0	rw	0xe000edb4: MPU Region Base Address Alias 3
MPU_RLAR_A3_NS	32	0	rw	0xe000edb8: MPU Region Limit Address Register Alias 3
MPU_MAIR0_NS	32	0	rw	0xe000edc0: MPU Memory Attribute Indirection Register 0
MPU_MAIR1_NS	32	0	rw	0xe000edc4: MPU Memory Attribute Indirection Register 1
FPCCR_NS	32	c0000000	rw	0xe000ef34: Floating Point Context Control
FPCAR_NS	32	0	rw	0xe000ef38: Floating Point Context Address
FPDSCR_NS	32	0	rw	0xe000ef3c: Floating Point Default Status Control

Table 13.5: Registers at level 1, group:System_non_secure

13.1.6 VFP

Registers at level:1, group:VFP

Name	Bits	Initial-Hex	RW	Description
d0	64	0	rw	
d1	64	0	rw	
d2	64	0	rw	
d3	64	0	rw	
d4	64	0	rw	
d5	64	0	rw	
d6	64	0	rw	
d7	64	0	rw	
d8	64	0	rw	
d9	64	0	rw	

d10	64	0	rw	
d11	64	0	rw	
d12	64	0	rw	
d13	64	0	rw	
d14	64	0	rw	
d15	64	0	rw	
FPSCR	32	0	rw	floating-point status/control

Table 13.6: Registers at level 1, group:VFP

13.1.7 Integration_support

Registers at level:1, group:Integration_support

Name	Bits	Initial-Hex	RW	Description
executionPri	32	7fffffff	r-	current execution priority level
stackDomain	64	7fff ecc6f020	r-	stack domain for current execution level
HaltReason	8	0	r-	bit field indicating halt reason
atomicType	8	0	r-	current atomic instruction type (1:atomic, 2:exclusive)
WakeWFINow	8	0	-w	wake up core from WFI state

Table 13.7: Registers at level 1, group:Integration_support